

Syed Nabil Shah

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EDUCATION

University of Leicester

Jun 2026

BSc Software Engineering - First Class Honours (80%, top 5% of cohort)

- **1st place** IBM/Capital One Hackathon. Built AidWay, a privacy-preserving humanitarian aid-allocation system.

RESEARCH

TandemRLVR - Scalable Oversight via Legible Reasoning

Ongoing

- Training reasoning agents to produce intermediate reasoning a weaker overseer can inspect, continue and **verify**, treating legibility as a measurable, **RLVR-trainable** property rather than an aesthetic one.
- Core loop: senior agent reasons partway through a task, deliberately weaker junior agent completes it, verifier scores the outcome; staged from heuristic placeholder agents to **real local LLM agents**.
- Built as reproducible RL research: modular agents, disciplined evaluation, diffable experiment outputs.

Cooperative AI - Emergence in Multi-Agent RL

2026

- Trained **PPO** agents (PyTorch, PettingZoo) under systematically varied reward structure.
- Found **selfish incentives produced more cooperation**: shared rewards blur credit assignment so no agent can isolate which action earned the reward, while selfish rewards keep the gradient signal sharp and cooperation emerges instrumentally.

EXPERIENCE

Minexx

Jul 2024 - Jan 2026

Software Engineer Intern

- Engineered async microservices for large-scale news scraping, translation & topic modeling; across multilingual sources (**3K+ articles/day**, **95% dedup rate**, **+20% sentiment accuracy**).
- Developed **Tin Sentiment Bot**, a production NLP system combining GDELT data ingestion, multilingual processing & PostgreSQL warehousing to issue real-time buy/sell/hold signals for the tin market.
- Built fault-tolerant **ETL** pipelines enabling **24/7 automated** data sync on Compute Engine VM.
- Migrated company AI systems to scalable GCP architecture, for a platform handling **1M+ requests/day**.
- **Tech**: Python, Asyncio, Pandas, PostgreSQL, Firebase, GCP, NLTK, JavaScript, Grok API

PROJECTS

AgentOps - Runtime Policy Enforcement for AI Agents

Present

- Built a runtime classification layer that authorises or blocks an agent's proposed actions before execution, combining hybrid RAG grounding, a deterministic rules engine and a human-in-the-loop approval gate (JWT + RBAC).
- Guaranteed exactly-once, auditable actions via a transactional outbox worker and a hash-chained, tamper-evident audit log; contained prompt injection through schema-validated tool use.
- Verified with **440 automated tests** and **6 hard-gated evaluation suites** (all passed, 0 regressions).
- **Tech**: Python, FastAPI, PostgreSQL, pgvector, TypeScript, Next.js, React, Docker, GitHub Actions, LangGraph

NoSzn - AI Copilot for DTC Shopify Brand

2026

- Shipped AI systems for a live Shopify clothing brand, syncing **4,200+ orders** and **62 SKU variants** into a 29-endpoint FastAPI + Next.js/TypeScript app with PII anonymisation and refund/discount reconciliation.
- Built a provider-agnostic LLM gateway with retries, rate limiting and streaming SSE/RAG copilot.
- Fine-tuned an open model (LoRA) on brand voice; added vision tagging, voice queries, token/cost tracking and NL-to-structured alerts; backed by 75 tests.
- **Tech**: Python, PostgreSQL, scikit-learn, Docker, Ollama, OpenAI API, Whisper, AWS Lambda, PEFT/LoRA

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, TypeScript/JavaScript, Bash

GenAI/LLMs: RAG, vector search, prompt engineering, LLM evaluation, guardrails, agentic workflows, Ollama

ML/Deep Learning: PyTorch, NumPy, TensorFlow, scikit-learn, Hugging Face, spaCy, NLTK, computer vision

Cloud & Data: GCP, AWS, Firebase, PostgreSQL, pgvector, Redis, Kafka

Tools & MLOps: FastAPI, Next.js, Docker, GitHub Actions (CI/CD), pytest, Git, Claude Code

AI Security: prompt injection defenses, schema-validated tool calling, audit logging, least-privilege agent design